Vehicle Insurance Management System

Abstract

The Vehicle Insurance Management System is a modern, secure, and scalable web-based application designed to automate and streamline the management of vehicle insurance services. Traditional insurance systems rely heavily on manual processes, resulting in inefficiencies, delayed claim settlements, lack of transparency, and poor customer experience. This project aims to address these challenges by developing a fully digital platform that manages the entire lifecycle of vehicle insurance policies.

The system enables customers to purchase vehicle insurance policies, view policy details, track renewals, and raise claims online. It also provides role-based functionalities for administrators, agents, and claims officers to efficiently manage policies, assist customers, and process claims. The application incorporates JWT-based authentication and role-based authorization to ensure secure access to system resources.

Developed using ASP.NET Core Web API for the backend ,Angular for the frontend and SQL Server for Database , the system follows clean architecture principles and implements RESTful services. Entity Framework Core is used for database operations, ensuring efficient data management. The platform also includes dashboards and reporting features that provide real-time insights into policy performance, claim status, and revenue generation.

Overall, the system enhances operational efficiency, reduces manual intervention, improves transparency, and delivers a seamless user experience, making it suitable for real-world insurance business environments.

Objectives

1. To develop a secure and scalable Vehicle Insurance Management System using ASP.NET Core, Angular and SQL Server.
2. To implement user authentication and role-based authorization using JWT.
3. To enable customers to:
    - Purchase vehicle insurance policies
    - View and manage policy details

- Raise and track insurance claims

4. To allow agents to:
   - Assist customers in policy creation
   - Manage policy data
5. To provide claims officers with:
   - Tools to review, approve, or reject claims
   - Efficient claim processing workflows
6. To enable administrators to:
   - Manage users, policies, and reports
   - Monitor system activities and performance
7. To implement policy lifecycle tracking:

   Draft → Active → Expired → Cancelled
8. To implement claim lifecycle management:

   Submitted → Under Review → Approved/Rejected → Settled
9. To provide dashboards and reports for better decision-making.
10. To ensure secure data handling, validation, and error management.